

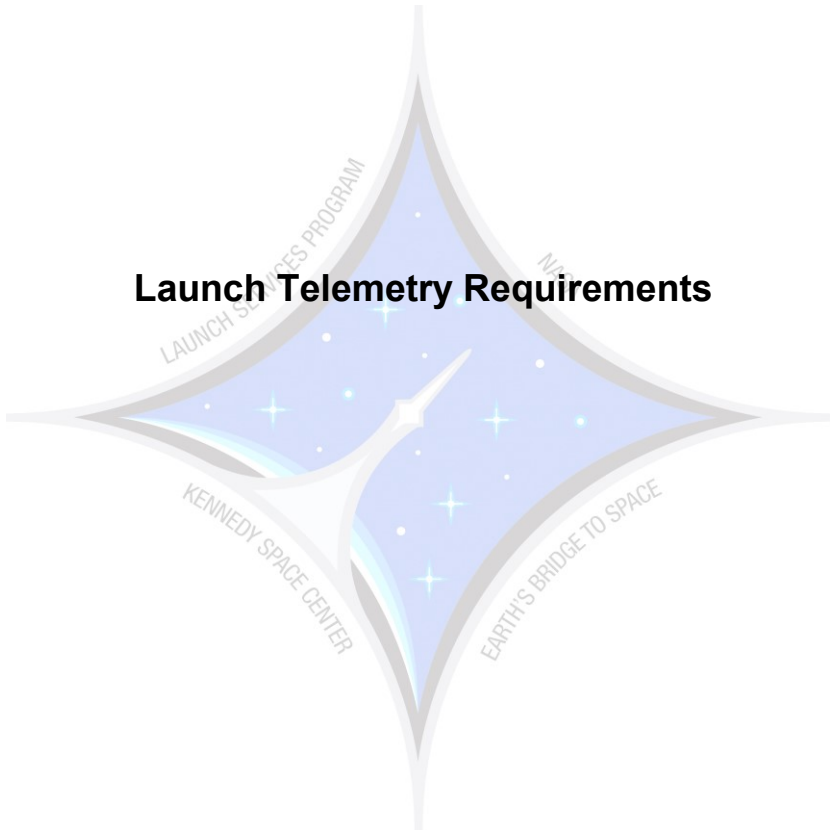


National Aeronautics and Space Administration  
John F. Kennedy Space Center, Florida

**LSP-PD-120.05**  
**Revision D**

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**Launch Services Program**

A large, stylized four-pointed star graphic with a blue and white gradient. The star has a central white point and is surrounded by a grey border. The text "Launch Telemetry Requirements" is centered over the star. The star's points are labeled with text: "LAUNCH SERVICES PROGRAM" at the top, "KENNEDY SPACE CENTER" at the bottom left, and "EARTH'S BRIDGE TO SPACE" at the bottom right. The star's interior is filled with a blue and white gradient and contains several small white stars.

## Launch Telemetry Requirements

Approved: Approved  
Jorge Piquero  
LSD&R Manager

Date: 02/03/2025

| Revision History |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Revision         | Description                                                                                                                                                                                                                                                                                                                                                                                                                          | Date           |
| Basic            | Basic release                                                                                                                                                                                                                                                                                                                                                                                                                        | April 27, 2005 |
| A                | Updated job titles & format, enumerated list of requirements, removed "may" from M2, added "for NASA primary SC missions" to R1, added R4, rephrased D3 & added D4 & *note regarding store & forward techniques to Desired, removed words regarding LSP Process Flow for Obtaining Launch Telemetry Support and AVO-format waiver from Waiver Authority, added **note regarding terminal count definition to definition of Mandatory | May 31, 2011   |
| B                | Added pre-launch requirements, changed definitions to highlight impact of loss of requirement                                                                                                                                                                                                                                                                                                                                        | June 5, 2015   |
| C                | Updated the Executive Summary on page 3, first sentence to clarify that this document is not applicable to VADR                                                                                                                                                                                                                                                                                                                      | June 9, 2023   |
| D                | Remove exception to store and forward telemetry coverage in the Table on page 3.                                                                                                                                                                                                                                                                                                                                                     | Feb 03, 2025   |

## EXECUTIVE SUMMARY:

This directive establishes the Launch Services Program policy for **minimum** launch vehicle telemetry support requirements for the day of launch for NLS II missions. This data is used prior to launch to validate launch vehicle readiness and to assist in anomaly resolution. This data is used during and after launch to verify mission success and spacecraft environmental contractual requirements.

| <b>Mandatory</b>                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1 - Receive and record all phases of powered flight utilizing the launch vehicle primary propulsion system prior to spacecraft separation for all NASA primary spacecraft missions. |
| M2 - Receive and record contamination and collision avoidance maneuvers that utilize the primary propulsion system for all NASA primary spacecraft missions.                         |
| <b>Required</b>                                                                                                                                                                      |
| R1 - Receive and record depletion burns for NASA primary spacecraft missions                                                                                                         |
| R2 - Receive and record all phases of powered flight utilizing the primary propulsion system prior to spacecraft separation for all NASA secondary spacecraft missions.              |
| R3 - Receive and record primary spacecraft separation.                                                                                                                               |
| R4 - Receive and record Launch Vehicle High Frequency Environmental Data System for NASA primary spacecraft missions                                                                 |
| R5 - Validate the status of the Mandatory and Required down range sites                                                                                                              |
| R6 - Receive prelaunch telemetry and voice at the NASA primary data station where the NASA vehicle engineering consoles are located.                                                 |

## **Definitions:**

**Required** – These requirements have minimal impact to mission success and mitigations are available if they are not satisfied. These requirements may be waived during terminal count by the NLM.

**Mandatory** – **Mandatory** – These requirements have an impact to the verification of mission success, and if not satisfied, mitigations are not available. Once within terminal count, a mandatory requirement cannot be waived. In exceptional cases, Mandatory requirements may be changed to Required by the Launch Services NLM prior to entering terminal count.  
**Secondary Spacecraft Mission** - This mission type will be identified in the Launch Service Provider contract.

**Terminal Count** begins at the start of dynamic launch day operations, such as propellant loading or aircraft movement.

**Note:** Waivers prior to the terminal count shall be identified in written or verbal format at the Flight Readiness Review or closest review thereafter. Waivers issued on launch day shall be announced verbally to the launch team by the NLM.